TomatoInsight - Data Tables

This document lists all data stored and used by the project.

Cloud storage is Firestore (plus Firebase Authentication). The browser also keeps a localStorage mirror for offline/fast UI.

Note: The conceptual ERD (User -> Plant -> Scan -> Disease Result, Subscription -> Payment) maps to the structures below. TomatoInsight uses farm tiles instead of a separate Plant table, and credits instead of a Subscription table.

Storage overview

Layer	Path / key	Purpose
Firebase Auth	Auth users	Login identity (uid, email, password)
Firestore	users/{uid}	Profile, credits, farm layout, plant names, week, scan calendar
Firestore	users/{uid}/tiles/{tileNum}	Saved scans, scores, week history, leaf photo
Firestore	users/{uid}/deletions/{id}	Deleted plant/tile history
Firestore	payments/{paymentId}	Demo credit purchases
Browser	localStorage	Local cache of the same farm/profile data

1. Firebase Auth - User account

Field	Type	Key	Description
uid	string	PK	Firebase Auth user id (also Firestore users document id)
email	string		Sign-in email
password	string		Handled by Firebase Auth (not stored in Firestore)
displayName	string		Optional display name from signup

Relationships

- 1 Auth user -> 1 Firestore users/{uid} document
- 1 Auth user -> many payments rows (uid)
- 1 Auth user -> many tiles and deletions subdocuments

2. Firestore - users (User / farmer account)

Document id: uid (PK)

Field	Type	Key	Description
uid	string	PK	Same as Firebase Auth uid (document id)
firstName	string		Given name
middleInitial	string		Single letter
lastName	string		Family name
farmName	string		Farm display name
email	string		Account email
photo	string		Profile photo (..., size-capped)

updatedAt	string (ISO)		Last profile update
credits	map		Credit balance and grant flags (see table 2a)
layout	map		Farm grid size (see table 2b)
selectedFarmWeek	number		Active farm week (4-12)
plantNames	map		Per-tile plant summary (see table 2c)
scanCalendar	map		Optional scan schedule (see table 2d)

Relationships

- 1 users -> many users/{uid}/tiles
- 1 users -> many users/{uid}/deletions
- 1 users -> many payments (via uid)

2a. Nested - users.credits

Field	Type	Description
balance	number	Current scan credits
signupGranted	boolean	true after one-time +10 signup credits
profileBonusGranted	boolean	true after one-time +5 farm-setup bonus
updatedAt	string (ISO)	Last credit change

2b. Nested - users.layout

Field	Type	Description
rows	number	Farm grid rows
columns	number	Farm grid columns
updatedAt	string (ISO) / number	Last layout save

2c. Nested - users.plantNames["{tileNum}"]

Field	Type	Description
name	string	Plant / tile name
diseaseLabel	string[]	Detected disease names
diseases	array	{ name, pct } (and related score fields)
healthy	map	Healthy score summary when present
recommendation	map	Treatment guide (see Intervention / recommendation)

2d. Nested - users.scanCalendar

Field	Type	Description
enabled	boolean	Custom scan schedule on/off
anchor	string (YYYY-MM-DD)	Schedule start date
nextDue	string (YYYY-MM-DD)	Next due scan date
selected	string (YYYY-MM-DD)	Selected scan day

3. Firestore - users/{uid}/tiles (Plant + Scan + Disease result)

Document id: tileNum (string, e.g. "1") (PK)

Parent FK: uid (from path)

Field	Type	Key	Description
tileNum	string/number	PK	Tile number (document id)
uid	string	FK	Owner (parent users/{uid})
plantName	string		Name for this plant/tile
savedAt	string (ISO)		Last save time
scanDayIso	string (YYYY-MM-DD)		Scan calendar day used
weekNumber	number		Farm week of latest save (4-12)
imageDataUrl	string / null		Latest leaf scan image (may be stripped if too large)
scores	map		Latest classification scores (see 3a)
topBarKey	string / null		Top class key (healthy, tlcv, early_blight, spotted_wilt)
topPct	number		Top class confidence %
weeks	map		Per-week history (see 3b)
recommendation	map		Latest treatment recommendation when synced

Relationships

- Many tiles -> 1 user
- 1 tile -> many week snapshots inside weeks
- 1 tile -> 0..1 recommendation (intervention guide)

3a. Nested - scores

Field	Type	Description
healthy	number	Healthy %
tlcv	number	Tomato leaf curl virus %
early_blight	number	Early blight %
spotted_wilt	number	Spotted wilt %

3b. Nested - weeks["{weekNumber}"] (Timeline / week history)

Field	Type	Description
weekNumber	number	Farm week
savedAt	string (ISO)	When that week was saved
scanDayIso	string	Scan day for that week
scores	map	Score snapshot
topBarKey	string / null	Top class for that week
topPct	number	Top % for that week
imageDataUrl	string / null	Week image (may be null to save space)

4. Recommendation / Intervention (guide)

Stored under:

- users/{uid}/tiles/{tileNum}.recommendation
- and/or users/{uid}.plantNames["{tileNum}"].recommendation
- and locally in tomatoInsight.plantRecommendations

Field	Type	Description
diseaseName	string	Disease this guide targets
urgencyNote	string	Short urgency note
summary	string	Treatment summary
actionSteps	string[]	Recommended actions
preventionTips	string[]	Prevention tips
isCriticalRemoval	boolean	Critical / remove-plant flag
barKey	string	Class key for the guide
pct	number	Related confidence %
topBarKey	string	Top disease key when saved
topPct	number	Top % when saved
scores	map	Score snapshot tied to the guide
scoresKey	string	Cache key for the score set
source	string	"groq" or "rules"
savedAt	string (ISO)	When the guide was saved
guides	array	Optional multi-disease guides

Relationships

- Many recommendations -> 1 tile / plant result

5. Firestore - users/{uid}/deletions (Deleted plant history)

Document id: deletion id (PK)

Parent FK: uid

Field	Type	Key	Description
id	string	PK	e.g. "{tileNum}-{timestamp}"
uid	string	FK	Owner (parent path)
tileNum	number		Tile that was cleared
plantName	string		Name at deletion time
deletedAt	string (ISO)		When deleted

Relationships

- Many deletions -> 1 user

6. Firestore - payments (Payment)

Document id: auto-generated paymentId (PK)

Field	Type	Key	Description
paymentId	string	PK	Firestore document id
uid	string	FK	Buyer (users/{uid})
email	string		Buyer email at purchase time
packId	string		starter / grower / season / custom
packName	string		Pack display name
credits	number		Credits added
listPriceUsd	number		Listed pack price

amountChargedUsd	number		Amount charged (demo = 0)
currency	string		e.g. USD
status	string		e.g. paid
method	string		e.g. demo
isDemo	boolean		true for demo checkout
createdAt	string (ISO)		Payment time
createdAtMs	number		Payment time (ms) for sorting
balanceBefore	number		Credit balance before top-up
balanceAfter	number		Credit balance after top-up

Relationships

- Many payments -> 1 user (uid)
- Payment updates users.credits.balance

Credit packs (catalog, not a DB table)

packId	packName	credits	listPriceUsd
starter	Starter	10	4.99
grower	Grower	25	9.99
season	Season	50	17.99

Entity relationship summary (as implemented)

```
Firebase Auth User (uid)
        |
        | 1:1
        v
   users/{uid}
   +-- credits (nested)
   +-- layout (nested)
   +-- plantNames (nested map of tiles)
   +-- scanCalendar (nested)
   +-- selectedFarmWeek
   |
   +-- 1:N --? users/{uid}/tiles/{tileNum}
   |              +-- scores / topBarKey / imageDataUrl
   |              +-- weeks{}          (timeline / week scans)
   |              +-- recommendation   (intervention)
   |
   +-- 1:N --? users/{uid}/deletions/{id}
   |
   +-- 1:N --? payments/{paymentId}
```

Mapping from conceptual ERD -> this project

Conceptual table	Actual storage
User	Firebase Auth + users/{uid}
Plant	users/{uid}/tiles/{tileNum} (+ plantNames)
Subscription	users.credits (balance + grant flags; packs via payments)
Payment	payments/{paymentId}
Scan	Tile latest fields + tiles.weeks
Disease Result	scores, topBarKey, topPct, plantNames.diseases
Intervention	recommendation on tile / plantNames

Timeline	weeks + selectedFarmWeek + scanCalendar
Notification	UI/local reminders (not a Firestore collection today)

7. Browser localStorage (cache / offline mirror)

Key	Mirrors / stores
tomatoInsight.userProfiles	Profile fields by uid
tomatoInsight.credits	Cached credit balance / flags by uid
tomatoInsight.farmLayout	Farm grid layout
tomatoInsight.tilePlantInfo	Tile scans / week history
tomatoInsight.plantNames	Plant names by tile
tomatoInsight.plantRecommendations	Recommendations by tile
tomatoInsight.selectedFarmWeek	Active week number
tomatoInsight.tileDeletions	Deletion history array
tomatoInsight.scanCalendar	Scan schedule
tomatoInsight.sidebarCollapsed	UI preference
Remember-me / toast flags	Auth remember + one-time credit toast flags

Local data is synced to Firestore when the user is signed in.

Credit rules (business data, not separate tables)

Event	Credits	One-time?	Stored flag
First account credit setup	+10	Yes	credits.signupGranted
Complete 6 farm setup steps + Update profile	+5	Yes	credits.profileBonusGranted
Successful Save Info (scan save)	-1	Per save	balance only
Demo pack purchase	+10 / +25 / +50	Per payment row	payments + balance